

AI Career Companion Project Documentation

Comprehensive Project Documentation
AI-Powered Career Guidance Platform

FastAPI	React 18	LangGraph	Groq LLM	SHAP
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Model: llama-3.1-8b-instant • Backend: Python / FastAPI • Frontend: React + Vite

1. Executive Summary

The AI Career Companion is a professional-grade, AI-powered Business Intelligence platform designed to transform raw spreadsheets into actionable insights, Machine Learning-ready datasets, and polished interactive dashboards.

It achieves this through a multi-module backend pipeline powered by FastAPI, cutting-edge machine learning libraries, and Groq's high-speed LLM inference engine, coupled with a premium React-based front-end executive dashboard.

Core value proposition

- Automated data understanding: schema detection, null normalisation, and LLM-powered column intelligence
- Intelligent preprocessing: statistical profiling, AI-generated cleaning plans, and versioned artifact management
- AI analyst interface: RAG-powered chatbot, dynamic chart generation, and parallel insight extraction
- Decision intelligence: AutoML, time-series forecasting, SHAP explainability, and LangGraph recommendation agents

2. System Architecture

The platform is organised into four sequential modules, each building on the outputs of the previous one. Raw data enters Module 1 and flows through the pipeline, ultimately surfacing as executive-level insights in the React dashboard.

Stage	Module	Primary responsibility
1	Data Understanding	Schema detection, null normalisation, LLM column intelligence, domain identification
2	Intelligent Preprocessing	Deep EDA, AI cleaning plan, rule engine, target detection, artifact versioning
3A	AI Analyst & Dashboard	RAG vector store, chart engine, insights engine, persona chatbot
3B	Decision Intelligence	AutoML, forecasting, SHAP/RCA, risk engine, LangGraph recommendations

3. Module Details

Module 1 — Data Understanding

The entry point of the pipeline. Normalises raw data, classifies column types automatically, and uses an LLM to generate human-readable explanations of cryptic column names and overall dataset meaning.

Module 1 — Feature Breakdown	
Establishes foundational data understanding before any transformation is applied.	
Null normalisation	Detects disguised nulls (empty strings, placeholder values) and standardises them for downstream processing.
Hybrid schema detection	Automatically classifies column data types — numeric, categorical, datetime, text — without manual annotation.
Column intelligence (LLM)	Queries Groq llama-3.1-8b-instant to explain cryptic or abbreviated column names in plain business language.
Domain detection	Identifies the business context of the dataset — Sales, HR, Finance, Logistics — to guide downstream module behaviour.
AI narrative summary	Generates a natural-language description of the entire dataset including its purpose, structure, and key characteristics.

Module 2 — Intelligent Preprocessing

Transforms raw, messy data into a clean, ML-ready dataset. Combines statistical profiling, an LLM-generated cleaning plan, a deterministic rule engine, and artifact versioning.

Module 2 — Feature Breakdown

Cleans, encodes, and scales data in a reproducible, AI-planned pipeline.

Deep EDA	Profiles outliers, skewness, correlations, and distributions across all columns to build a comprehensive statistical picture.
Agent planner (Groq)	Generates a structured JSON data cleaning plan based on EDA results, specifying which operations to apply and in what order.
Rule engine	Safely executes cleaning tasks (imputation, encoding, scaling, outlier handling) in a canonical, reproducible order.
Target detection	Automatically identifies potential target labels or prediction columns for downstream ML tasks.
Artifact manager	Version-controls and persists trained encoders, scalers, and preprocessing artifacts for reproducibility and reuse.

Module 3A — AI Analyst & Dashboard

Gives users a natural language interface to their data. Combines a retrieval-augmented chatbot, a dynamic chart engine, and parallel insight generation into a single analyst experience.

Module 3A — Feature Breakdown

Surfaces insights and charts through an LLM-powered conversational interface.

RAG vector store	Ingests EDA reports and module summaries into a retrieval store so the chatbot can ground its answers in real dataset facts.
Chart engine (LLM-driven)	The LLM dynamically plans 5-6 Plotly-compatible business charts appropriate to the dataset's domain and structure.
Parallel insights engine	Runs multiple insight-generation threads simultaneously to surface trends, anomalies, and KPI summaries instantly.
Persona chatbot	Lets business users converse with their data in plain language — no SQL or technical jargon required.

Module 3B — Decision Intelligence

A standalone predictive and prescriptive analytics engine. Runs AutoML, time-series forecasting, root cause analysis, and risk monitoring, then synthesises results into board-level recommendations via a LangGraph agent.

Module 3B — Feature Breakdown

Converts cleaned data into predictions, forecasts, and actionable business strategies.

AutoML prediction engine	Automatically selects and tunes classification or regression models (XGBoost, LightGBM, CatBoost) to fit the dataset.
Forecasting engine	Uses Facebook Prophet and PMDArima for time-series forecasting with confidence intervals and seasonality decomposition.

Root cause analysis (SHAP)	Integrates SHAP to explain model predictions, surfacing which features drive each outcome for business stakeholders.
Risk & drift engine	Monitors incoming data for statistical anomalies and distribution drift to flag data quality or model degradation risks.
LangGraph recommendation agent	An autonomous agent pipeline that synthesises outputs from all engines into prioritised, actionable business strategy recommendations.

4. Technology Stack

Backend Infrastructure

Category	Libraries / Frameworks
API Framework	FastAPI, Uvicorn (high-performance async API server)
Data Processing	Pandas, NumPy, SciPy
Machine Learning	Scikit-learn, XGBoost, LightGBM, CatBoost
Time Series	Prophet (Facebook), PMDArima (ARIMA)
Explainability	SHAP, LIME
AI Orchestration	LangChain, LangGraph
Validation	Pydantic v2

Frontend Ecosystem

Category	Libraries / Frameworks
Framework	React 18+ with Vite build system
State Management	Zustand (pipeline and data state tracking)
Visualisations	Recharts (bar, line, composition charts)
Styling	TailwindCSS with Vanilla CSS (glassmorphism, vibrant palettes)
Animations	Framer Motion (micro-animations)
Icons	Lucide React

LLM Integration

Groq Inference Engine Primary model: llama-3.1-8b-instant

Used across all four modules: column intelligence (M1), cleaning plan generation (M2), chart and insight synthesis (M3A), and decision synthesis (M3B). Strict JSON prompting ensures deterministic, Pydantic-validated API contracts between AI components and the backend.

5. Project History & Evolution

The project evolved through four targeted phases to build its robust 4-module architecture.

Phase 1	Module 1 & 2 — Foundation Established the core data understanding pipeline (schema detection, semantic meaning via LLM) and intelligent preprocessing (deep EDA, anomaly detection, Groq-powered JSON cleaning plan, dataset preparation).
Phase 2	Module 3A — AI Analyst Integration Introduced the AI Analyst component. Implemented RAG vector stores over EDA summaries, a dynamic Plotly-compatible chart engine, a parallel insights engine, and a persona-based chatbot for natural language data conversations.
Phase 3	Module 3B — Decision Intelligence Developed a standalone engine handling AutoML prediction, Prophet/ARIMA forecasting, SHAP-powered Root Cause Analysis, anomaly and drift detection, and a LangGraph agent pipeline synthesising actionable recommendations.
Phase 4	Unified Dashboard — React Frontend Built the executive-level React interface with Zustand state management, Framer Motion micro-animations, Recharts visualisations, and a glassmorphism TailwindCSS layout for high-level business strategy presentation.